

CampusWasteGuide

AI-Powered Campus Waste Segregation Chatbot

Project Submission Report

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SDG Alignment: SDG 12 (Primary) · SDG 11 · SDG 13 · SDG 4 (Secondary)

1. Problem Statement

Problem Statement (Guideline Format)

How might we use AI to guide campus users in correctly segregating waste so that campuses can become more sustainable and reduce landfill contribution?

Context & Evidence

Campus waste bins are routinely contaminated — recyclables, compostable food waste, and hazardous e-waste are all mixed into general waste bins due to a lack of accessible, real-time guidance. Bin labels are static and cannot handle ambiguous questions such as "Is a coffee cup recyclable?" or "Where does a pen drive go?". E-waste items such as batteries, chargers, and USB drives are discarded in general bins, causing environmental and health hazards. Campus sustainability audits globally report 40–60% waste misclassification rates. No interactive, on-demand, accessible guidance tool exists for students and staff at the actual point of disposal.

Who is Affected?

Students (especially new intakes), lecturers, administrative staff, canteen and facility workers, and campus sustainability teams are all directly affected. The broader environment — through increased landfill, contaminated recyclables, and improper e-waste disposal — is the ultimate stakeholder.

Why is AI Needed?

AI enables real-time information retrieval at scale, handling natural-language queries that static labels cannot. A conversational AI assistant can answer ambiguous questions instantly, operate 24/7 without human staff, and provide consistent, accurate guidance to every user regardless of literacy level or prior knowledge.

2. Detailed Solution Description

CampusWasteGuide is a conversational AI chatbot that accepts a natural-language description of any waste item and instantly returns the correct disposal bin — drawn from a verified, 200-item campus waste knowledge base covering all 5 bin categories. The solution is implemented as a Python CLI application that dynamically loads its knowledge base from a plain-text file at runtime, requiring no API keys, no internet connection, and no external dependencies.

RAG Pipeline — 3 Stages

Stage 1 — Knowledge Base: knowledge_base.txt serves as the retrieval corpus. It contains 200 numbered waste items mapped to 5 bin destinations, organised into 5 human-readable categories. Any authorised staff member can update the rules without coding knowledge.

Stage 2 — Retrieval: The user's input is normalised (lowercased, parenthetical qualifiers stripped) and matched against all 200 knowledge base entries using bidirectional substring matching — the query may contain the keyword or vice versa. This handles natural language variation without requiring an LLM.

Stage 3 — Generation: A friendly, plain-English response is returned naming the exact bin. If no match is found, a polite fallback message directs the user to physical bin labels — ensuring no incorrect guidance is ever given.

Knowledge Base Summary

Blue Recycling Bin: 40 items (bottles, cans, paper, cardboard, cartons, metal)

Green Compost Bin: 40 items (food scraps, fruit/veg, garden waste, natural wood)

Black General Waste: 52 items (wrappers, foam, broken items, non-recyclables)

Admin Office E-Waste Drop-off: 40 items (devices, cables, bulbs, batteries, networking)

Red Sanitary Waste Bin: 28 items (hygiene, PPE, diapers, medical kits)

Total: 200 items across 5 destinations

3. AI Elements and Tools Used

Retrieval-Augmented Generation (RAG): The core architecture. knowledge_base.txt is the retrieval corpus; no LLM hallucination is possible because answers are always retrieved from verified rules, never generated from a language model's weights.

Conversational AI Interface: A natural-language input loop accepts free-text queries. No fixed command syntax is required — users type in plain English as they would to a human assistant.

IBM BOB: Used throughout the project for design, coding, knowledge base curation, analysis scripting, documentation, and presentation generation. Prompt engineering workflows were used to curate all 200 knowledge base entries systematically.

Prompt Engineering: Knowledge base rules were authored using structured prompt workflows. Item classification logic and edge cases (e.g. food-soiled vs clean paper) were resolved through iterative prompting.

Python 3: waste_assistant.py implements the dynamic KB loader (regex parser), normalisation layer, keyword matcher, and interactive CLI loop.
analyse_knowledge_base.py provides automated quality assurance — numbering checks, duplicate detection, coverage gap analysis, and 10-case query simulation.

4. Prototype & Demo

The working prototype is a fully functional Python CLI chatbot. To run it:

python waste_assistant.py

Sample Interactions (Verified Output)

Input: banana peel → Output: Green Compost Bin

Input: old laptop charger → Output: Admin Office E-Waste Drop-off

Input: chips packet → Output: Black General Waste

Input: sanitary pad → Output: Red Sanitary Waste Bin

Input: milk carton → Output: Blue Recycling Bin

Input: broken umbrella → Output: Black General Waste

Input: smart watch → Output: Admin Office E-Waste Drop-off

Input: leftover rice → Output: Green Compost Bin

Input: styrofoam cup → Output: Black General Waste

Input: used tissue → Output: Red Sanitary Waste Bin

Analysis & Test Results: analyse_knowledge_base.py was run and confirmed:

- 200 items across 5 categories — numbering continuous 1 to 200, no gaps
- 0 duplicate item names
- 0 coverage gaps between knowledge_base.txt and waste_assistant.py
- 10 / 10 simulated query tests PASSED

Demo Links

Local Demo: Run python waste_assistant.py from the project directory

No internet connection, API key, or paid service is required to run the prototype.

5. Responsible AI Considerations

Fairness: No bias in data or assumptions. All campus users receive identical, consistent guidance regardless of who asks — no personalisation, no discrimination.

Transparency: All rules are stored in a human-readable `knowledge_base.txt` file that any staff member can open, read, and verify. There is no black-box model involved.

Ethics: No harmful, discriminatory, or misleading outputs. The chatbot only answers waste disposal questions. When uncertain, a polite fallback prevents incorrect guidance from being given.

Privacy: Zero personal data is collected. No login, no tracking, no analytics, no user profiling of any kind. Usage is completely anonymous.

Human Oversight: The fallback response always directs uncertain users to physical bin labels. AI never fabricates an answer when the item is not found in the knowledge base.

Explainability: Every response names the exact bin, not just a colour code. Users understand the specific destination, making the guidance actionable and verifiable.

6. Expected Impact

What changes if this solution is implemented?

Waste misclassification drops at the point of disposal — guidance is instant and always available, unlike posters that are often ignored or absent. Recyclable diversion from general waste increases, delivering a measurable sustainability impact on campus recycling rates. E-waste reaches the proper Admin Office drop-off instead of landfill, reducing toxic contamination of soil and groundwater. Students build long-term habits through repeated correct interactions, creating a cultural shift rather than a one-time campaign. Campus sustainability KPIs improve without costly infrastructure changes — the solution requires only a text file and a Python interpreter.

Scalability Roadmap

Phase 1 (Current): CLI chatbot — fully functional, zero dependencies, runs locally

Phase 2: Web chatbot — Flask/FastAPI deployment, accessible via campus intranet browser

Phase 3: Messaging bot — WhatsApp/Telegram integration for mobile accessibility

Phase 4: LLM-backed version — semantic search with IBM Granite models for natural language understanding beyond keyword matching

7. Project Files

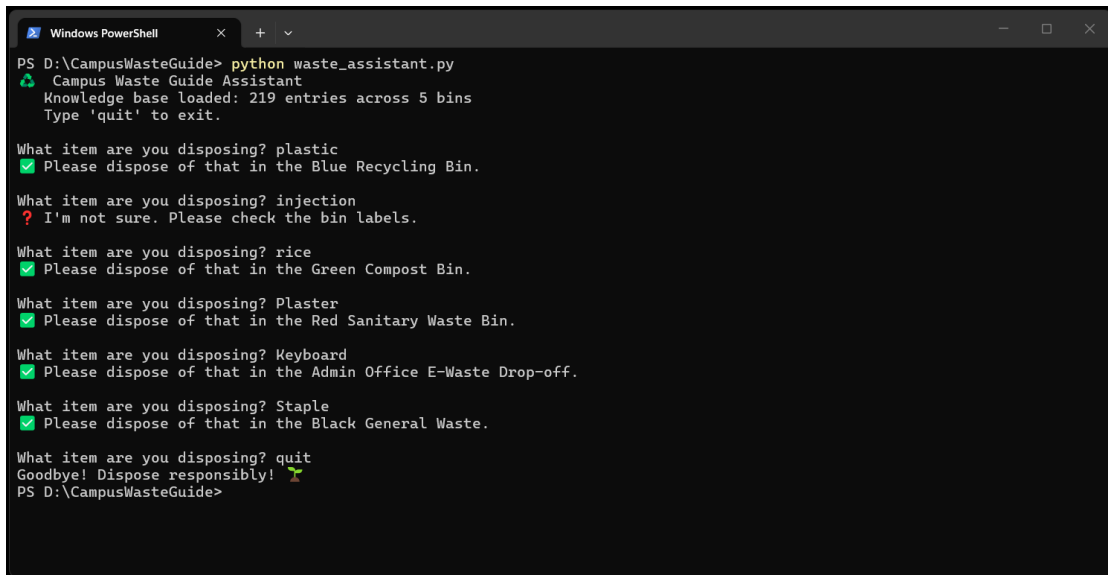
knowledge_base.txt — 200-item waste segregation knowledge base (5 categories, plain text)

waste_assistant.py — CLI chatbot with dynamic KB loader, regex parser, keyword matcher

analyse_knowledge_base.py — automated QA and test script (10/10 tests passed)

CampusWasteGuide_Submission.pptx — 11-slide guidelines-compliant submission deck

7. Screenshot



```
PS D:\CampusWasteGuide> python waste_assistant.py
Campus Waste Guide Assistant
Knowledge base loaded: 219 entries across 5 bins
Type 'quit' to exit.

What item are you disposing? plastic
✔ Please dispose of that in the Blue Recycling Bin.

What item are you disposing? injection
? I'm not sure. Please check the bin labels.

What item are you disposing? rice
✔ Please dispose of that in the Green Compost Bin.

What item are you disposing? Plaster
✔ Please dispose of that in the Red Sanitary Waste Bin.

What item are you disposing? Keyboard
✔ Please dispose of that in the Admin Office E-Waste Drop-off.

What item are you disposing? Staple
✔ Please dispose of that in the Black General Waste.

What item are you disposing? quit
Goodbye! Dispose responsibly! ♻️
PS D:\CampusWasteGuide>
```